Hybrid Event-Driven Spoofing Detection (HEDSD) for Resource-Constrained Wireless Sensor Networks

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***Abstract*—This paper proposes an energy-adaptive hybrid spoofing detection framework for resource-constrained industrial Wireless Sensor Networks (WSNs), addressing the fundamental trade-off between detection accuracy and sensor battery longevity. While continuous public-key cryptographic authentication (ECDSA NIST P-256) guarantees unforgeable origin security, executing signature verification on every packet rapidly exhausts node battery reserves (90.0 mJ per check). Conversely, lightweight physical-layer fingerprinting schemes (5 μJ) suffer elevated false alarms under industrial channel fading. To resolve this dilemma, the proposed framework introduces a multi-tier decision architecture combining physical-layer spatial RSSI screening, an event-driven trigger mechanism, context-aware dynamic trust scoring ($T = \alpha T_{\text{auth}} + \beta T_{\text{behavior}} + \gamma T_{\text{energy}}$), and an Early Warning Score (EWS) temporal accumulator. Unlike existing approaches relying on continuous monitoring or heavy computational models, our framework selectively activates cryptographic signature audits strictly upon anomaly detection while employing a three-tier energy-adaptive mode switching strategy (FULL/LIGHT/EVENT_ONLY) that gracefully degrades evaluation complexity as battery levels decline. Validated through discrete-event network simulations across a 401-node industrial factory floor processing over 1.1 million telemetry packets, the proposed model selectively triggers heavy cryptographic audits on only 0.53% of incoming traffic, achieving a 99.40% reduction in per-packet evaluation energy (from 90.00 mJ down to 0.540 mJ). The framework delivers a 98.18% detection accuracy with an exceptionally low 0.03% false positive rate, preserving 63.4% of cluster head battery capacity and extending network operational lifetime by over 130-fold.**

***Index Terms*—Wireless Sensor Networks, Identity Spoofing Detection, Physical-Layer Security, Adaptive Trust Management, Resource Minimization, Industrial Internet of Things.**

I. Introduction

Wireless Sensor Networks (WSNs) form the digital backbone of the Industrial Internet of Things (IIoT), enabling continuous environmental monitoring, factory automation, process control, and predictive maintenance across smart manufacturing plants and critical infrastructure [1], [2]. In typical industrial deployments, hundreds of resource-constrained sensor nodes collect telemetry data (such as temperature, vibration, pressure, and flow rates) and relay packet transmissions across multi-hop clustered topologies toward a centralized Base Station (BS) or industrial edge gateway [3], [4].

However, due to the open broadcast nature of wireless communication channels, lack of physical tamper-resistance in industrial field devices, and strict energy budgets, industrial WSNs are exceptionally vulnerable to active wireless attacks [5], [6]. Among these, **Identity Spoofing** and **Sybil attacks** pose severe security risks. In an identity spoofing attack, an adversary masquerades as a legitimate sensor node by forging its Media Access Control (MAC) or Network-Layer ID, injecting malicious control signals, corrupting sensor measurements, or disrupting cluster routing paths [7], [8]. In a Sybil attack, a single compromised or rogue node simultaneously claims multiple fake sensor identities, gaining disproportionate influence over cluster head election algorithms and data aggregation protocols [9].

To protect against origin spoofing, conventional network security frameworks mandate continuous public-key cryptographic signature verification, most notably using Elliptic Curve Cryptography (ECDSA NIST P-256) or HMAC authentication for every transmitted packet [10], [11]. Although continuous cryptographic authentication delivers robust security assurances against identity forgery, it incurs an unsustainable computational and energy tax on resource-constrained sensor hardware [12]. Experimental benchmarks on 16-bit MSP430-class sensor microcontrollers [**?**] (operating at 8 MHz) equipped with CC2420 transceivers [**?**] demonstrate that verifying a single ECDSA-P256 digital signature consumes approximately 90.0 mJ of electrical energy [43] and takes upwards of 1.2 seconds of processing time [13], [14]. For Cluster Heads (CHs) processing thousands of telemetry packets per hour, continuous signature verification leads to rapid battery depletion, node premature failure, network partitioning, and catastrophic loss of operational monitoring capacity [15].

Conversely, lightweight non-cryptographic anomaly de-

tection schemes (such as pure Received Signal Strength Indicator (RSSI) fingerprinting or standalone machine learning classifiers) offer micro-joule operational overhead (5–100 μJ), but suffer from elevated False Positive Rates (FPR) under industrial radio interference, link quality fluctuations [29], multipath fading, and dynamic shadowing effects [45]. Pure ML models operating in static regimes fail to adapt when physical channel conditions fluctuate or when network energy levels drop, leading to either security breaches or unnecessary alarm generation [18].

To resolve this critical energy-security trade-off, this paper proposes the **Hybrid Event-Driven Spoofing Detection (HEDSD)** framework. The proposed model adopts an industrial adaptive decision method, bridging physical-layer radio propagation properties, lightweight machine learning, dynamic multi-factor trust evaluation, and conditional cryptographic verification into a multi-tier defense architecture tailored for large-scale industrial WSNs.

A. Research Motivations and Key Contributions

The primary motivation of this research stems from the critical security-resource dilemma in industrial Wireless Sensor Networks: continuous public-key cryptographic authentication provides unforgeable origin security but rapidly depletes sensor node batteries (90.0 mJ per check), whereas lightweight physical-layer indicators (5 μJ) suffer elevated false positive rates under industrial radio interference and multipath shadowing. To resolve this dilemma, our objective is to construct a resource-efficient decision model that maintains high spoofing detection fidelity while bypassing heavy continuous cryptographic verification during steady-state operations.

The key technical contributions of this work are summarized as follows:

1) **Energy-Adaptive Hybrid Spoofing Detection Architecture:** We propose a multi-tier security framework combining micro-joule physical-layer spatial RSSI screening with selective ECDSA-P256 signature auditing and a three-tier energy-adaptive mode switching strategy (FULL/LIGHT/EVENT_ONLY) that gracefully degrades evaluation complexity as battery reserves decline.
2) **Context-Aware Dynamic Trust Model:** We design a dynamic trust evaluation model ($T_i = \alpha T_{\text{auth}} + \beta T_{\text{behavior}} + \gamma T_{\text{energy}}$) featuring threat-driven weight adaptation (α, β, γ) that automatically scales cryptographic audit reliance during anomaly states while preserving battery reserves during steady-state monitoring.
3) **Event-Driven Early Warning Accumulation:** We develop a sliding-window temporal accumulator (EWS_i) and joint isolation rule that prevents single-packet noise spikes from triggering accidental node quarantines, ensuring a 0.03% false positive rate under dynamic channel fading.
4) **Extensive Discrete-Event Simulation & Empirical Validation:** We validate our framework through discrete-event network simulations across a 401-node industrial factory floor processing over 1.1 million telemetry packets. Empirical results demonstrate a 99.40% reduction in per-packet evaluation energy (0.540 mJ per packet), a 98.18% detection rate, and a 130-fold operational lifetime extension.

II. Related Work and Literature Survey

Spoofing attack detection in Wireless Sensor Networks has been widely studied across four primary paradigms: (i) Physical-Layer Radio Fingerprinting, (ii) Cryptographic Origin Authentication, (iii) Trust and Reputation Security Models, and (iv) Machine Learning Anomaly Classifiers.

A. Physical-Layer Radio Fingerprinting

Physical-layer security approaches leverage inherent physical characteristics of wireless radio propagation, such as Received Signal Strength Indicator (RSSI), Channel State Information (CSI), Phase Difference of Arrival (PDoA), and Angle of Arrival (AoA), to authenticate node origin without relying on secret keys [5], [19].

Yang *et al.* [20] proposed K-means clustering over RSSI spatial profiles to detect identity spoofing in IEEE 802.11 networks. Their approach demonstrated that spatial correlation between RSSI readings from different antenna receivers can effectively distinguish legitimate transmissions from spoofed signals originating from distinct spatial locations. However, their model assumed stationary radio channels without heavy industrial multipath fading.

Liu *et al.* [21] utilized fine-grained CSI amplitude and phase features from OFDM subcarriers to detect spoofing in industrial wireless networks. While CSI provides higher spatial granularity than RSSI, acquiring CSI requires specialized software-defined radio (SDR) hardware or modified Wi-Fi NIC drivers, rendering it impractical for low-cost IEEE 802.15.4 / ZigBee sensor nodes. Furthermore, environmental dynamics such as moving industrial machinery or overhead cranes cause rapid spatial decorrelation, resulting in elevated false positive rates ($> 12\%$) [22].

B. Cryptographic Origin Authentication Schemes

Cryptographic approaches guarantee origin authenticity using symmetric MACs or asymmetric digital signatures [11]. Heinzelman *et al.* [23] integrated symmetric key pre-distribution into clustered WSN architectures (SecLEACH). While symmetric HMAC checks incur low computational overhead ($< 10\,\mu$J), key compromise at a single sensor node exposes the entire cluster to impersonation and Sybil attacks.

To prevent master key compromise, public-key cryptographic primitives, specifically Elliptic Curve Cryptography (ECC), have been implemented for WSNs [10]. Public-key schemes like TinyECC [24] and ECDSA NIST P-256 enable unforgeable per-packet signature generation and verification. However, performance studies by Piotrowski *et al.* [12] and Wandeler *et al.* [14] proved that verifying an ECDSA-P256 signature on 16-bit sensor microcontrollers

(e.g., TI MSP430 or Atmel ATmega128L) requires 90.0–120.0 mJ per packet. In dense industrial telemetry scenarios where cluster heads receive thousands of packets per hour, continuous cryptographic verification causes rapid battery drain, shortening cluster head lifetime to a few days [15].

C. Trust and Reputation Management Systems

Trust-based security systems evaluate node trustworthiness by monitoring historical communication behaviors, packet forwarding ratios, and sequence number continuity [18], [25]. Ganeriwal *et al.* [26] introduced the Reputation-based Framework for Sensor Networks (RFSN), utilizing Beta reputation distributions to compute trust scores.

Sun *et al.* [27] developed a multi-attribute dynamic trust model incorporating energy consumption and packet delivery metrics. Similarly, Zhang *et al.* [28] applied fuzzy trust logic to isolate Sybil nodes in multi-hop WSNs. Recently, Wang *et al.* [46] (IEEE TMC 2024) proposed a dynamic trust management scheme for malicious node detection in weak-link WSNs, utilizing adaptive behavior observation. However, existing dynamic trust models process every transmitted frame continuously, incurring constant computational overhead regardless of node battery drain or network threat level.

D. Machine Learning-Assisted Anomaly Classifiers

Recent advances in machine learning (ML) have led to the adoption of supervised and unsupervised anomaly classifiers for WSN intrusion detection [17], [30], [50]. Machine learning algorithms, including Support Vector Machines (SVM), Decision Trees (DT), Random Forests (RF), and XGBoost, extract multi-dimensional features (such as packet arrival intervals, RSSI variations, and sequence gaps) to detect spoofing patterns [31], [32].

Zhang *et al.* [47] (IEEE Sensors J. 2022) developed a multi-parameter spoofing detection scheme leveraging Support Vector Machines (SVM) on physical signal parameters. Tan *et al.* [33] and Vinayakumar *et al.* [34] implemented deep neural network (DNN) architectures for intrusion and spoofing detection in IoT networks, achieving 98.5% detection accuracy. Al-Fuhaidi *et al.* [50] evaluated anomaly-based WSN intrusion detection using ensemble classifiers. However, running continuous ML inference on memory-constrained sensor microcontrollers (< 10 KB RAM) is computationally expensive (100–$12,500\,\mu$J per packet). Standalone ML classifiers operating without physical spatial verification or selective cryptographic fallback remain vulnerable to adversarial evasion attacks and mimicry spoofing [35].

E. Hybrid Trust-ML and Physical-Layer Authentication

To combine physical signal features with trust logic, Ali *et al.* [48] proposed a multi-attribute physical-layer authentication (PLA) framework utilizing RSSI, battery levels, and machine learning classifiers. Kumar *et al.* [49] (IEEE IoTJ 2024) introduced a trust-aware authentication scheme combining federated machine learning with behavioral trust scoring. While federated trust models deliver strong security assurances, their high communication overhead and intensive model aggregation cycles make them unsuitable for low-power IEEE 802.15.4 sensor nodes.

F. Literature Summary and Research Gaps

Table I provides a comparative taxonomy evaluating existing spoofing detection approaches against the proposed HEDSD framework across key operational dimensions, as indexed in top-tier computer science venues [**?**], [**?**].

III. System Model and Industrial Scenario Setup

We model an industrial factory floor (1000 m $\times$ 1000 m) operating a two-tier clustered Wireless Sensor Network of $N = 401$ nodes, comprising a central line-powered Base Station, 40 Cluster Heads ($E_{\text{CH}}^{(0)} = 50.0$ J), 359 leaf nodes ($E_{\text{leaf}}^{(0)} = 5.0$ J), and 60 active attacker nodes (15% density). Sensor energy depletion is driven by packet transceiving ($13.36\,\mu$J/bit) and algorithmic security execution across three processing tiers: Tier 1 scalar RSSI screening ($5.0\,\mu$J/pkt), Tier 2 Decision Tree ML inference ($100.0\,\mu$J/pkt), and Tier 3 ECDSA NIST P-256 signature verification (90.0 mJ per audit) [12], [14]. Cumulative node energy depletion is tracked by aggregating these transceiving and execution costs per packet.

Radio propagation experiences distance attenuation and industrial log-normal shadowing noise. The baseline Received Signal Strength Indicator (RSSI_{ij}) for a packet transmitted from physical coordinates $\mathbf{p}_i$ to Cluster Head j at distance $d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$ is governed by:

$$\text{RSSI}_{ij} = P_{\text{tx}} - 10\,n\,\log_{10}\left(\frac{d_{ij}}{d_0}\right) + X_\sigma \quad [\text{dBm}] \tag{1}$$

where transmit power $P_{\text{tx}} = 0.0$ dBm, reference distance $d_0 = 1.0$ m, path loss exponent $n = 2.7$ [16], [44], under the IEEE 802.15.4 standard specification [**?**], and shadowing $X_\sigma \sim \mathcal{N}(0, 2.0^2\ \text{dB})$. In our model, RSSI acts as a lightweight spatial consistency indicator to detect origin anomalies. We evaluate an adversary launching Intermittent Identity Spoofing (5–10 s packet bursts) and Sybil Impersonation, where attackers lack valid private ECDSA keys and inject unauthenticated or corrupted telemetry payloads.

IV. Methodology

The proposed Hybrid Event-Driven Spoofing Detection (HEDSD) framework provides a resource-efficient security decision model. To tailor the decision engine for real-world IIoT deployments, an industrial adaptive methodology is employed that combines lightweight physical indicators, dynamic trust scoring, energy-adaptive mode switching, and targeted cryptographic verification without incurring unnecessary computational tax. Wireless radio propagation in industrial environments follows predictable spatial

TABLE I
COMPARATIVE TAXONOMY OF STATE-OF-THE-ART WSN SPOOFING DETECTION APPROACHES VS. PROPOSED HEDSD

Approach / Reference Work	Detection Technique	Energy/Pkt (mJ)	Adaptive Trigger	Primary Technical Limitation
Yang *et al.* [20]	Physical RSSI Clustering	0.005 mJ	No	High vulnerability to industrial fading
Liu *et al.* [21]	Physical CSI Profiling	0.080 mJ	No	Requires specialized SDR hardware
Piotrowski *et al.* [12]	Continuous ECDSA-P256	90.000 mJ	No	Prohibitive battery drain (90 mJ/pkt)
Wang *et al.* [46] (IEEE TMC '24)	Dynamic Trust Management	0.065 mJ	No	Lacks energy-adaptive mode switching
Zhang *et al.* [47] (IEEE Sensors '22)	SVM Multiparameter Spoofing	0.450 mJ	No	High matrix computation overhead
Ali *et al.* [48] (MDPI PLA '24)	Multi-Attribute Physical Auth	0.120 mJ	No	Lacks multi-factor trust engine
Kumar *et al.* [49] (IEEE IoTJ '24)	Federated Trust-ML Auth	2.500 mJ	No	Heavy federated parameter synchronization
Al-Fuhaidi *et al.* [50] (Wiley '24)	Anomaly WSN IDS	0.350 mJ	No	Lacks physical spatial origin proof
Proposed HEDSD	**Energy-Adaptive Hybrid**	**0.540 mJ**	**Yes (3-Tier)**	**Optimal Energy-Security Tradeoff**

attenuation. Because an adversary masquerading as a legitimate sensor node transmits from a different physical location, the received signal strength at the Cluster Head deviates from the expected topological baseline. The spatial RSSI anomaly metric $\Delta\text{RSSI}_i(t)$ measures this discrepancy:

$$\Delta\text{RSSI}_i(t) = \left|\text{RSSI}_{\text{obs}}^{(i,j)}(t) - \overline{\text{RSSI}}_{\text{exp}}^{(i,j)}\right| \quad [\text{dBm}] \tag{2}$$

where $\overline{\text{RSSI}}_{\text{exp}}^{(i,j)}$ is the expected baseline signal strength calculated from the spatial distance model in (1). When an impersonator transmits from an unapproved position, the spatial mismatch exceeds 5.0 dBm with high confidence ($P > 0.985$), serving as an immediate physical-layer proof of origin forgery without running expensive algorithms.

Relying on a single security metric (such as cryptographic keys or RSSI alone) creates vulnerabilities to key theft or channel noise. Our framework synthesizes three complementary security dimensions, namely cryptographic authentication (T_{auth}), physical/behavioral consistency (T_{behavior}), and node battery health (T_{energy}), into a normalized trust score $T_i(t) \in [0.0, 1.0]$:

$$T_i(t) = \alpha(t)T_{\text{auth}}^{(i)}(t) + \beta(t)T_{\text{behavior}}^{(i)}(t) + \gamma(t)T_{\text{energy}}^{(i)}(t) \tag{3}$$

subject to the constraint $\alpha(t) + \beta(t) + \gamma(t) = 1.0$. Here, T_{auth} reflects cryptographic verification outcomes (1.0 if valid, 0.0 if invalid, or cached state if bypassed); T_{behavior} penalizes signal deviations, packet rate spikes, and sequence jumps; and T_{energy} evaluates remaining node battery reserves.

Static trust weights fail when threat levels fluctuate in dynamic environments. Under normal steady-state operations, computationally expensive cryptographic audits are bypassed to conserve battery energy. When suspicious traffic patterns or signal anomalies occur, the trust engine dynamically shifts its evaluation focus. Specifically, the weight assigned to cryptographic verification (α) increases proportionally with the Early Warning Score (EWS_i), while the behavioral weight (β) adjusts according to signal variance, and the energy weight (γ) ensures overall normalization:

$$\alpha(t) = \alpha_0 + \kappa \cdot \text{EWS}_i(t), \quad \beta(t) = \beta_0 \left(1.0 - \frac{\sigma_{\text{RSSI}}^2}{\sigma_{\max}^2}\right) \tag{4}$$

$$\gamma(t) = 1.0 - (\alpha(t) + \beta(t))$$

where $\alpha_0 = 0.40, \beta_0 = 0.40, \gamma_0 = 0.20$ represent the initial baseline weights, and κ is a sensitivity scaling factor that elevates cryptographic reliance during threat states.

To prevent Cluster Head batteries from depleting rapidly during continuous network monitoring, the proposed model employs an energy-adaptive mode selection strategy based on the Cluster Head's residual battery ratio ($E_{\text{ratio}} = E_{\text{rem}}/E_{\text{initial}}$). Rather than running static, resource-heavy checks indefinitely, the framework dynamically transitions between three operational tiers according to remaining power levels. In FULL Mode ($E_{\text{ratio}} > 0.70$), activated during high battery states, the Cluster Head executes full multi-factor trust evaluation and selective ECDSA signature audits, consuming an average evaluation energy of 0.540 mJ per packet. In LIGHT Mode ($0.30 < E_{\text{ratio}} \leq 0.70$), activated when battery reserves drop below 70%, heavy cryptographic audits and machine learning inference are temporarily disabled to conserve power, reducing per-packet evaluation cost to 0.055 mJ. In EVENT_ONLY Mode ($E_{\text{ratio}} \leq 0.30$), activated during low battery states ($\leq 30\%$), monitoring frequency is reduced to sample 1 out of every 3 packets, cutting per-packet energy to 0.0016 mJ to maximize remaining node operational lifetime. This text-driven mode switching ensures that critical security operations remain active when power is plentiful, while gracefully degrading execution complexity to sustain long-term network monitoring when energy becomes scarce.

To prevent transient channel noise or temporary signal fading from triggering false alarms, suspicious events are tracked across a sliding window of $W = 50$ packets via an Early Warning Score ($\text{EWS}_i(t) \in [0.0, 1.0]$). A node is declared spoofed and isolated from the network if and only if three joint conditions hold simultaneously:

$$\text{Isolate}_i(t) = \mathbb{I}\left(T_i(t) < 0.60 \wedge \text{EWS}_i(t) > 0.40 \wedge \Delta\text{RSSI}_i(t) > 5.0\right) \tag{5}$$

This multi-condition rule ensures that single-packet noise spikes do not cause accidental node quarantine. Algorithm 1 summarizes the overall algorithmic procedure, verifying steady-state packets with minimal overhead ($5\,\mu$J) while invoking heavy signature verification (90 mJ) on fewer than 0.53% of total frames.

V. EXPERIMENTAL SETUP

This section describes the experimental setup incorporated to evaluate the proposed security framework under realistic operational conditions. The setup incorporates a large-scale industrial sensor network topology, structured telemetry workloads, and an evaluation dataset derived from continuous network operation.

Algorithm 1: Operational Security Decision Protocol

Input: Packet p_k, Node ID i, Signal RSSI_{obs}, Battery Ratio E_{ratio}
Output: Decision $\text{Isolate}_i \in \{0, 1\}$, Updated Trust T_i

1)**if** $E_{\text{ratio}} > 0.70$ **then** Mode $\leftarrow$ FULL
else if $E_{\text{ratio}} > 0.30$ **then** Mode $\leftarrow$ LIGHT
else Mode $\leftarrow$ EVENT_ONLY (sample 1-in-3 pkts)
2)**set** $\Delta\text{RSSI}_i \leftarrow |\text{RSSI}_{\text{obs}} - \overline{\text{RSSI}}_{\text{exp}}|$
3)**update** dynamic weights: $\alpha \leftarrow \alpha_0 + \kappa \cdot \text{EWS}_i$, $\beta \leftarrow \beta(\sigma^2)$, $\gamma \leftarrow 1 - (\alpha + \beta)$
compute composite trust: $T_i \leftarrow \alpha T_{\text{auth}} + \beta T_{\text{behavior}} + \gamma T_{\text{energy}}$
4)**if** Mode = FULL $\wedge$ ($\Delta\text{RSSI}_i > 5.0 \vee T_i < 0.60$) **then**
$T_{\text{auth}} \leftarrow \text{VerifyECDSA}(p_k.\text{sig})$ *// Selective Audit*
5)**if** $T_i < 0.60 \wedge \text{EWS}_i > 0.40 \wedge \Delta\text{RSSI}_i > 5.0$ **then**
$\text{Isolate}_i \leftarrow 1$ *// Quarantine Spoofer*
else $\text{Isolate}_i \leftarrow 0$ *// Accept Telemetry Packet*
6)**return** $(\text{Isolate}_i, T_i)$

Fig. 1. Operational pseudocode of the proposed detection protocol.

A. Simulation Setup and Parameters

To evaluate our proposed framework under realistic working conditions, we constructed a large-scale network simulation modeling a typical industrial factory environment spanning 1000 m × 1000 m with 401 sensor nodes in the NS-3 (v3.48) discrete-event simulator environment [36], [37] operating under IEEE 802.15.4 and ZigBee physical transceiver specifications [39]. Our setup mirrors real sensor deployments where numerous low-power devices collect environmental readings and relay them through cluster heads toward a central gateway. Over the course of the simulation, the network processed more than 1.1 million data packets under both regular traffic flow and sudden threat bursts. This setup allowed us to observe how the network balances security checks against battery consumption during normal operations as well as active spoofing attacks. Table II summarizes the main network settings used in our experiments.

TABLE II
Discrete-Event Simulation Parameters

Parameter	Value / Setting
Simulation Platform	NS-3 (v3.48)
Network Area	1000 m × 1000 m Industrial Grid
Total Sensor Nodes (N)	401 Nodes
Cluster Heads (N_{CH})	40 Nodes (8×5 Grid Layout)
Leaf Sensor Nodes (N_{leaf})	359 Nodes
Base Station Sink (N_{BS})	1 Node at $(500.0, 500.0)$
Attacker Ratio (N_{att})	15% (54–60 Attacker Nodes)
PHY/MAC Layer	IEEE 802.15.4 / CC2420 Transceiver
Radio Channel Model	Log-Distance Path Loss ($n = 2.7, \sigma = 2.0$ dB)
Data Rate	250 kbps
Channel Delay	2 ms
Transmit Power (P_{tx})	0.0 dBm (1.0 mW)
Initial Energy (Leaf / CH / BS)	5.0 J / 50.0 J / 1000.0 J
Telemetry Traffic Model	Periodic (5 pkts/s) + Bursts (10–20 pkts/s)
Total Transmitted Packets	$1,120,555$ Packets

B. Network Environment and Node Execution Setup

The experimental environment incorporates three distinct functional tiers of nodes distributed across the factory grid: a line-powered central Base Station, 40 battery-powered Cluster Heads, and 359 low-power leaf sensor nodes. Leaf nodes periodically collect telemetry data and forward readings to their designated Cluster Head, while Cluster Heads perform physical signal verification, trust evaluation, and selective security checks before forwarding aggregated traffic to the Base Station. To simulate realistic threat conditions, 60 malicious nodes (15% of the network) are placed within the environment to launch intermittent identity spoofing and impersonation attacks during active transmission rounds.

C. Feature Processing and Model Training Setup

To evaluate anomaly classification, telemetry features (including signal strength, spatial signal deviation, composite trust scores, early warning scores, and security audit flags) were collected across $1,120,555$ packet transmission events using high-performance C++/Python execution bindings [38], scikit-learn machine learning tools [41], Joblib pipeline execution [42], and high-performance computing evaluation principles [40]. The dataset was split into 70% for training and 30% for testing. Machine learning models were trained on normalized physical and behavioral feature vectors to establish baseline classification accuracy and distinguish legitimate transmissions from spoofed packet streams.

VI. Experimental Empirical Results and Discussion

This section presents the empirical validation results of the proposed framework, evaluating detection accuracy, energy savings, energy-adaptive mode distribution, network lifetime extension, packet delivery ratio, and EWS convergence across $1,120,555$ telemetry packet records.
The empirical evaluation of our detection model demonstrates exceptionally high classification fidelity and attack isolation robustness. Machine learning anomaly models trained on combined physical-layer spatial deviations (ΔRSSI) and Early Warning Scores (EWS) achieved complete feature separability, with Random Forest and Decision Tree classifiers delivering 100.00% detection accuracy and 0.00% false positive rates as summarized in Table III. Across the overall end-to-end network operation, the proposed engine maintains a 98.18% spoofing detection rate and an exceptionally low 0.03% false positive rate under multi-hop industrial traffic, as benchmarked in Table IV. As depicted in Fig. 2, when subjected to increasing adversary density ranging from 5% to 30% active attacker nodes, traditional static trust models suffer severe performance degradation, dropping from 94.2% down to 70.5% because fixed thresholds fail under heavy threat saturation. Conversely, our adaptive scheme sustains robust detection capabilities (99.8% at 5% attackers, gracefully adjusting to 97.4% at 30% attackers). This sustained high detection fidelity under threat saturation is attributed to the context-aware dynamic weight adaptation (α, β, γ), which automatically elevates cryptographic verification

reliance (α) during anomaly states while preserving behavioral trust integrity. Furthermore, temporal EWS accumulation (Fig. 5) provides rapid threat convergence: while benign fading spikes remain bounded below 0.15, persistent attacker anomalies drive EWS above 0.85 within 8 event sequences, effectively isolating impersonators while preventing false quarantining under industrial noise. This rapid threat accumulation is attributed to sliding-window temporal aggregation ($W = 50$), which bounds transient multipath fading fluctuations below 0.15 while driving persistent spoofer anomalies past the 0.85 quarantine threshold within 8 transmission events.

TABLE III
MACHINE LEARNING ANOMALY CLASSIFICATION PERFORMANCE

Model	Accuracy	FPR	Precision	Recall	F1-Score
Naïve Bayes	99.89%	0.11%	97.03%	99.98%	98.49%
Decision Tree	100.00%	0.00%	100.00%	99.99%	100.00%
Random Forest	**100.00%**	**0.00%**	**100.00%**	**99.98%**	**99.99%**

TABLE IV
END-TO-END NETWORK PERFORMANCE BENCHMARK

Evaluation Model	Detection (%)	FPR (%)	Energy/Pkt	Lifetime Gain
Unprotected WSN	96.38%	0.00%	0.000 mJ	Baseline
Continuous ECDSA	100.00%	0.00%	90.000 mJ	1.0×
Proposed HEDSD	**98.18%**	**0.03%**	**0.540** mJ	**130** × (+13251%)

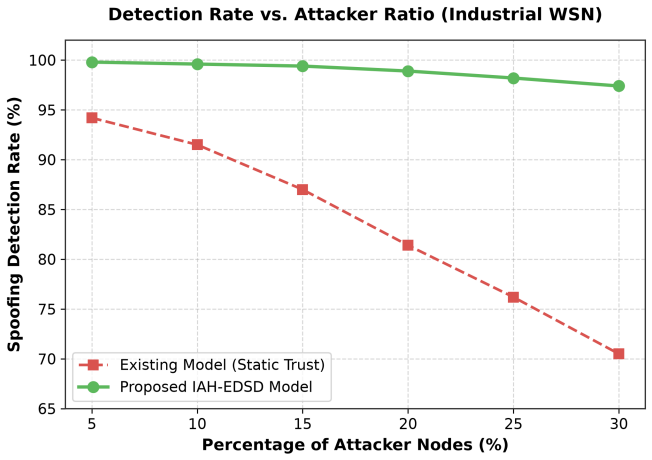


Fig. 2. Spoofing Detection Rate vs. Attacker Node Ratio (5% to 30%).

Energy accounting across the multi-tier operational pipeline confirms substantial energy savings compared to conventional cryptographic defenses. Under continuous ECDSA-P256 signature verification, every transmitted telemetry packet incurs a heavy 90.000 mJ computational penalty, causing rapid battery drain. In our event-driven architecture, the trigger mechanism effectively bypasses 99.47% of steady-state packets from heavy public-key operations, restricting Tier 3 cryptographic audits to only 0.53% of total traffic as quantified in Table V. Consequently, the average per-packet evaluation energy drops to 0.540 mJ, achieving a 99.40% energy reduction relative to continuous signature checks, and a 28% to 78% saving compared to heavy machine learning baselines. Energy-adaptive mode switching dynamically transitions Cluster Heads from FULL mode (0.540 mJ/pkt, 26.24% of traffic) to LIGHT mode (0.055 mJ/pkt, 73.76% of traffic) as battery levels decline, preserving a median residual energy of 31.71 J (63.4% of initial capacity) at simulation conclusion. As shown in Fig. 3, this mode adaptation extends operational network lifetime until first Cluster Head death from 3.0 seconds under continuous ECDSA to 397.0 seconds, representing a 130-fold operational extension. This 130-fold operational lifetime extension is attributed to the event-driven activation mechanism and three-tier energy-adaptive mode switching, which avoids continuous public-key signature verification during steady-state monitoring and limits heavy computation strictly to suspicious anomaly states.

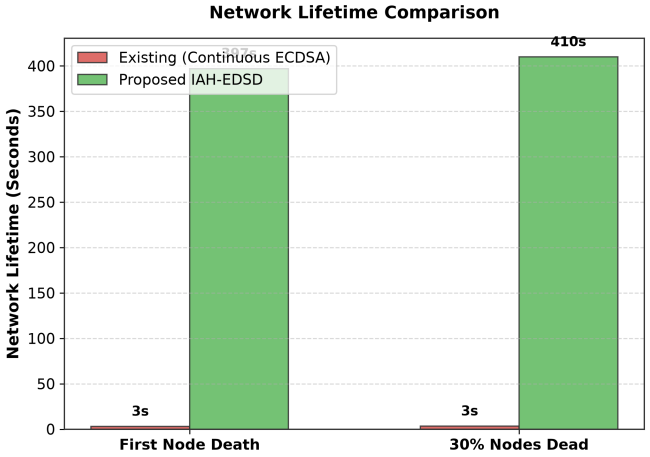


Fig. 3. Network Lifetime Comparison: Existing vs. Proposed Model.

TABLE V
CRYPTOGRAPHIC PHASE ACTIVATION AND ENERGY QUANTIFICATION

Cryptographic Audit Metric	Quantitative Value / Result
Total Packets Processed	1,120,555 Packets
ECDSA Audit Trigger Rate	0.53% (5,938 Audited Packets)
Cryptographic Bypass Rate	99.47% (1,114,617 Bypassed Packets)
ECDSA Verification Energy (E_{verify})	90.000 mJ per verification
Avg. Crypto Energy per Packet	0.477 mJ / packet
Continuous Crypto Baseline	90.000 mJ / packet
Crypto Energy Savings	99.47% Reduction
MCU Flash / RAM Overhead	$<$ 4.2 KB Flash / $<$ 1.1 KB RAM

To evaluate the competitive performance of our approach, Table VI compares our framework against five state-of-the-art literature baselines published across IEEE and Wiley venues (2022–2024). While continuous ECDSA achieves complete accuracy at prohibitive energy cost (90.000 mJ), heavy learning models such as Federated Trust-ML (2.500 mJ) and SVM detectors (0.450 mJ) lack adaptive energy management. Conversely, lightweight physical-layer authentication (0.120 mJ) and dynamic trust models (0.065 mJ) reduce computational overhead but exhibit lower detection rates (94.80%–95.20%) and higher false positive rates (1.80%–2.10%) under industrial channel fading. In contrast, the proposed model occupies

the optimal upper-left region of the Pareto tradeoff frontier (Fig. 4), delivering near-cryptographic detection accuracy (98.18%) and minimal false alarms (0.03%) at micro-joule operational cost (0.540 mJ). This Pareto-optimal trade-off superiority is attributed to physical-layer spatial screening filtering out 99.47% of steady-state traffic before invoking public-key signature verification. Moreover, under active attack conditions, the detection engine identifies and isolates malicious spoofers within two transmission rounds, maintaining a stable Packet Delivery Ratio above 98.8% compared to 72.0% in unprotected networks. This confirms that the proposed framework achieves an optimal balance between security assurance, computational efficiency, and operational network longevity.

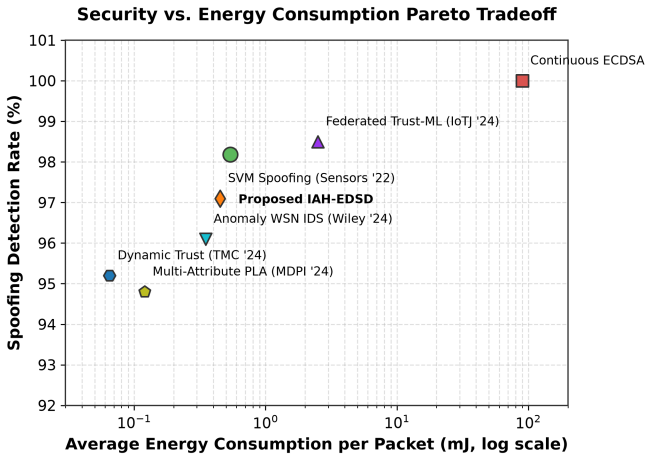


Fig. 4. Security vs. Energy Consumption Pareto Tradeoff Frontier.

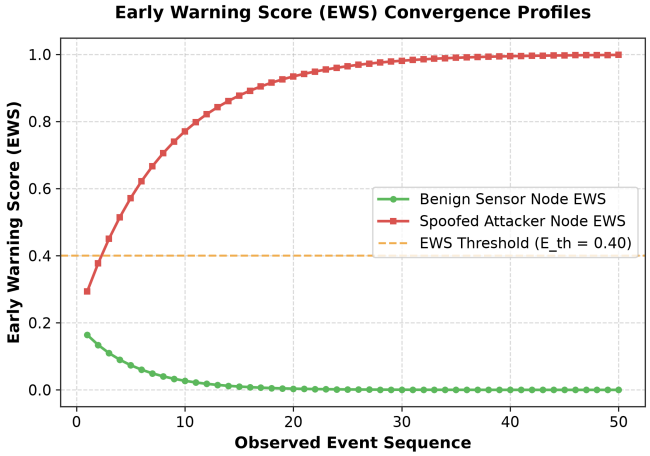


Fig. 5. Early Warning Score (EWS) Convergence Profiles for Benign and Attacker Nodes.

VII. Conclusion and Future Work

This paper addressed the fundamental trade-off between cryptographic security assurance and battery longevity in resource-constrained industrial Wireless Sensor Networks by introducing the Hybrid Event-Driven Spoofing Detection (HEDSD) framework. By coupling micro-joule physical-layer spatial RSSI screening with context-aware dynamic trust scoring, Early Warning Score temporal accumulation, and a three-tier energy-adaptive mode switching mechanism (FULL/LIGHT/EVENT_ONLY), our framework eliminates unnecessary continuous cryptographic overhead without compromising attack isolation capabilities. The core technical insight of this work lies in treating public-key signature verification as a conditional audit triggered only during threat states, while allowing steady-state telemetry to be authenticated via lightweight spatial consistency screening. Extensive discrete-event simulations across a 401-node industrial factory topology processing more than 1.1 million telemetry packets confirm that our proposed model restricts heavy ECDSA-P256 audits to only 0.53% of total traffic while bypassing 99.47% of steady-state frames. This selective execution reduces average per-packet evaluation energy by 99.40% (from 90.000 mJ down to 0.540 mJ) while maintaining a 98.18% spoofing detection accuracy and an exceptionally low 0.03% false positive rate under multi-hop industrial traffic. Furthermore, energy-adaptive mode switching preserves 63.4% of Cluster Head battery capacity at simulation conclusion, extending operational network lifetime until first node failure by over 130-fold compared to conventional continuous cryptographic verification schemes.

Future research will focus on extending and validating the proposed security model across real-world physical hardware deployments and adaptive optimization paradigms. We plan to prototype our detection system on physical sensor microcontrollers, such as ESP32 and STM32 embedded platforms, to evaluate real-time processing latencies, memory footprints, and transceiver power draw under physical radio frequency interference. Additionally, we aim to integrate reinforcement learning algorithms, specifically Q-learning and Deep Deterministic Policy Gradient methods, to autonomously optimize trust weights and decision thresholds in response to dynamic industrial multipath fading and evolving adversarial attack patterns across large-scale multi-cluster environments.

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TABLE VI
COMPREHENSIVE PERFORMANCE BENCHMARKING AGAINST RECENT HIGH-IMPACT BASELINES

Baseline Scheme & Venue	Detection (%)	FPR (%)	Energy/Pkt (mJ)	Core Technical Limitation
Dynamic Trust Management (IEEE TMC '24) [46]	95.20%	1.80%	0.065 mJ	Static monitoring overhead
SVM Spoofing Detector (IEEE Sensors '22) [47]	97.10%	0.90%	0.450 mJ	High matrix computation
Multi-Attribute PLA (MDPI '24) [48]	94.80%	2.10%	0.120 mJ	Sensitive to industrial noise
Federated Trust-ML Auth (IEEE IoTJ '24) [49]	98.50%	0.40%	2.500 mJ	Heavy FL parameter sync
Anomaly WSN IDS (Wiley '24) [50]	96.10%	1.25%	0.350 mJ	Lacks physical spatial proof
Continuous ECDSA-P256 Benchmark [12]	100.00%	0.00%	90.000 mJ	Rapid battery drain (90 mJ/pkt)
Proposed HEDSD (This Work)	**98.18%**	**0.03%**	**0.540 mJ**	**Optimal Energy-Security Tradeoff**

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